

PROJECT: Custom VPC setup with Bastion Host & NAT Gateway

Services: VPC, EC2, NAT Gateway, Bastion Host, Security Groups.

Project Goal: Build a secure private/public subnet architecture with internet access for private instances.

Skills: Routing Tables, Subnetting, SSH Tunnelling.

Definition: A custom VPC setup with a Bastion Host and NAT Gateway is an AWS network architecture where a user-defined Virtual Private Cloud is created with public and private subnets, a Bastion Host is deployed in the public subnet to provide secure administrative access to private instances, and a NAT Gateway is used to allow private instances outbound internet access without exposing them to inbound internet traffic.

Scope of the Project: The scope of this project is to design and implement a secure and scalable AWS network architecture using a custom Virtual Private Cloud (VPC) that separates resources into public and private subnets.

The project focuses on providing:

- Secure administrative access to provide EC2 instance using a Bastion Host.
- Outbound internet connectivity for private instances using a NAT Gateway.
- Strong network isolation and access control using security groups and route tables.

Methodologies:

1. Bastion Host and NAT Gateway: Method used in this Project.

- A Bastion Host is placed in the public subnet.
- All private EC2 instances are placed in private subnets.
- SSH access is allowed:
 - Local Machine: Bastion Host
 - Bastion Host: Private EC2
- NAT Gateway allows private EC2 instances to access the internet only for outbound traffic.

2. Direct Public EC2 Access:

- All EC2 instances have Public Ips.
- SSH directly from local machine.
- Its Insecure.
- Exposes instances to internet.
- Not recommended for production

3. AWS System Manager (SSM) Session Manager Method:

- Access private EC2 via AWS console.
- No Bastion Host required.
- Very secure.
- It requires IAM roles & SSM agent.
- Less networking exposure.

Multiple access methods can be used to design a secure AWS network, including Direct Public Access, SSM Session Manager, Bastion Host and NAT Gateway. Among these, the Bastion Host and NAT Gateway method is preferred because it provides,

- Strong Security
- Controlled Access
- Simplicity and real-world applicability while keeping costs manageable.

Services Used in this Project:**1. Amazon VPC (Virtual Private Cloud)**

- **Definition:** A logically isolated virtual network in AWS where users can define IP ranges, subnets, and routing.
- **Purpose:** To create a secure and customizable network environment.
- **Role in this project:** Hosts all networking components including subnets, route tables, and gateways.

2. Amazon EC2 (Elastic Compute Cloud)

- **Definition:** A web service that provides scalable virtual servers in the cloud.
- **Purpose:** To run applications and system services.
- **Role in this project:** Used to host the Bastion Host and Private EC2 instances.

3. Bastion Host

- **Definition:** A hardened EC2 instance deployed in a public subnet to securely access private resources.
- **Purpose:** To provide controlled administrative access.
- **Role in this project:** Acts as the single secure entry point for SSH access to provide EC2 instances.

4. NAT Gateway (Network Address Translation Gateway)

- **Definition:** A managed AWS service that allows private subnet instances to access the internet without inbound exposure.
- **Purpose:** To enable outbound internet connectivity.
- **Role in this project:** Allows private EC2 instances to download updates and access external services securely.

5. Internet Gateway (IGW)

- **Definition:** A gateway that connects a VPC to the public internet
- **Purpose:** To enable internet access for public resources.
- **Role in this project:** Provides internet connectivity to the Bastion Host and NAT Gateway.

6. Subnets (Public and Private)

- **Definition:** Logical network segments within a VPC.
- **Purpose:** To isolate resources based on access requirements.
- **Role in this project:** Public subnet hosts the Bastion Host and NAT Gateway, while Private subnet hosts internal EC2 instances.

7. Route Table

- **Definition:** A set of rules that controls network traffic routing within a VPC.
- **Purpose:** To direct traffic to appropriate destinations.
- **Role in this project:** Routes public subnet traffic to IGW and private subnet traffic to NAT Gateway.

8. Security Groups

- **Definition:** Virtual firewall that control inbound and outbound traffic for AWS resources.
- **Purpose:** To enforce network-level security.
- **Role in this project:** Restricts SSH access to private EC2 instances only through the Bastion Host.

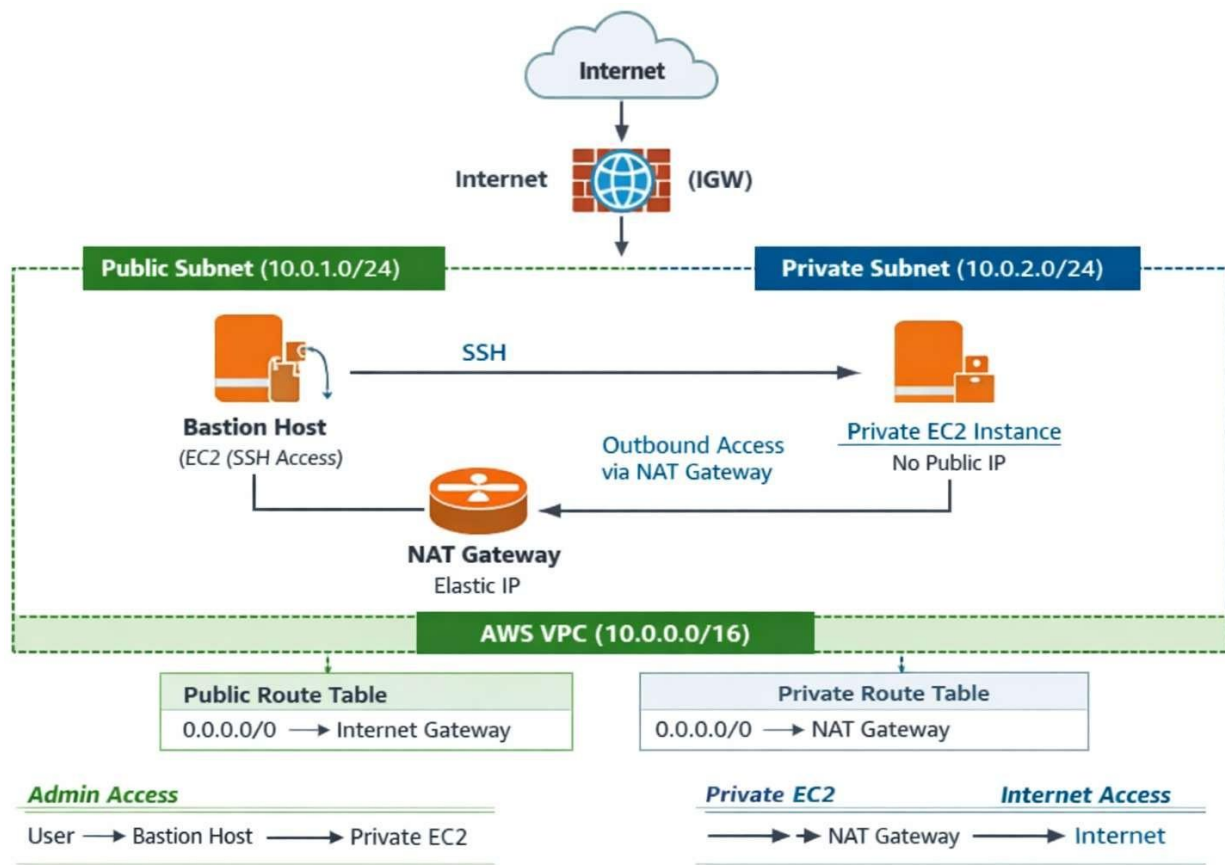
9. Elastic IP (EIP)

- **Definition:** A static public IPv4 address designed for dynamic cloud computing.
- **Purpose:** To provide a stable public IP.
- **Role in this project:** Assigned to the NAT Gateway to enable consistent outbound internet access.

10. SSH (Secure Shell)

- **Definition:** A cryptographic network protocol for secure remote login.
- **Purpose:** To securely access and manage servers.
- **Role in this project:** Used to connect from the local machine to the Bastion Host and from the Bastion Host to the private EC2.

Architecture Diagram



Implementation and Execution of a custom VPC setup with Bastion Host and NAT Gateway.

Phase 1: Create VPC.

- Open AWS Management Console → VPC
- Click create VPC
- Select VPC only
- Configure:
 - Name: Bastion-VPC
 - IPV4 CIDR: 10.0.0.0/16
 - Tenancy: Default
- Click create VPC

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the user's name 'Pavan (2074-5447-8223)'. The left sidebar shows the 'VPC dashboard' with a list of VPCs. The main content area displays the details for the VPC 'vpc-0369b1ff18be76af3 / Bastion-VPC'. A green banner at the top of the details section states 'You successfully created vpc-0369b1ff18be76af3 / Bastion-VPC'. The details are organized into a grid:

Details		Info	
VPC ID	vpc-0369b1ff18be76af3	State	Available
DNS resolution	Enabled	Tenancy	default
Main network ACL	acl-0a71cc5f579368099	Default VPC	No
IPv6 CIDR (Network border group)	-	Network Address Usage metrics	Disabled
Encryption control ID	-	Encryption control mode	-
Block Public Access	Off	DNS hostnames	Disabled
DHCP option set	dopt-06208c8207636ba54	Main route table	rtb-0053f5a42de4903e9
IPv4 CIDR	10.0.0.0/16	IPv6 pool	-
Route 53 Resolver DNS Firewall rule groups	-	Owner ID	207454478223

Phase 2: Create Subnets

Public Subnet:

- Go to Subnets → Create Subnet
- Select VPC: Bastion-VPC
- Subnet Name: Bastion-Public-Subnet1
- Availability Zone: ap-south-1a
- IPV4 CIDR: 10.0.1.0/24
- Click create subnet.

Private Subnet:

- Go to Subnets → Create Subnet
- Subnet Name: Bastion-Private-Subnet1
- Availability Zone: ap-south-1a
- IPV4 CIDR: 10.0.2.0/24
- Click create Subnet

The screenshot shows the AWS Management Console interface for the VPC dashboard. A green notification banner at the top states: "You have successfully created 4 subnets: subnet-0fca43674a45a84b1, subnet-0186390bbe5e1a0ca, subnet-03f1d40bd7ce0d1fb, subnet-0236b0e51bc911552". Below this, the "Subnets (4)" section is active, displaying a table of the created subnets.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
Private-Subnet-2	subnet-0236b0e51bc911552	Available	vpc-0d89618c353d7f13d Scal...	Off	10.0.4.0/24
Public-Subnet-1	subnet-0fca43674a45a84b1	Available	vpc-0d89618c353d7f13d Scal...	Off	10.0.1.0/24
Private-Subnet-1	subnet-03f1d40bd7ce0d1fb	Available	vpc-0d89618c353d7f13d Scal...	Off	10.0.3.0/24
Public-Subnet-2	subnet-0186390bbe5e1a0ca	Available	vpc-0d89618c353d7f13d Scal...	Off	10.0.2.0/24

Phase 3: Create Internet Gateway (IGW)

- Go to Internet Gateway → Create
- Name: Bastion-IGW
- Click create Internet Gateway
- Select IGW → Actions → Attach to VPC
- Choose Bastion-VPC
- Click on Attach.

The screenshot shows the AWS Management Console interface for the Internet Gateway (IGW) details page. A green notification banner at the top states: "Internet gateway igw-0609f1b7ed9cb7c86 successfully attached to vpc-0d89618c353d7f13d". Below this, the "igw-0609f1b7ed9cb7c86 / Web-IGW" section is active, displaying details for the attached gateway.

Internet gateway ID	State	VPC ID	Owner
igw-0609f1b7ed9cb7c86	Attached	vpc-0d89618c353d7f13d Scalable-Web-VPC	207454478223

Phase 4: Create Public Route Table

Public Route Table:

- Go to Route Tables → Create route table
- Name: Bastion-Public-RT1
- VPC: Bastion-VPC
- Click create route table

Add Route:

- Select Bastion-Public-RT1
- Routes: Edit routes
- Add routes:
 - Destination: 0.0.0.0/0
 - Target: Internet Gateway (Bastion-IGW)
- Save routes

The screenshot shows the AWS Management Console interface for the 'Public-RT' route table (rtb-0abb1fa2dc8ed4546). A green notification banner at the top states 'Updated routes for rtb-0abb1fa2dc8ed4546 / Public-RT successfully'. The left sidebar shows the 'Virtual private cloud' section with 'Route tables' selected. The main content area displays the 'Details' tab for the route table, showing its ID, VPC, and owner ID. Below this, the 'Routes' tab is active, showing a table of routes. The table has columns for Destination, Target, Status, Propagated, and Route Origin. Two routes are listed: one for 0.0.0.0/0 targeting 'igw-0609f1b7ed9cb7c86' with an 'Active' status, and another for 10.0.0.0/16 targeting 'local' with an 'Active' status. The 'Route Origin' for the first route is 'Create Route' and for the second is 'Create Route Table'.

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0609f1b7ed9cb7c86	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Associate Public Subnet:

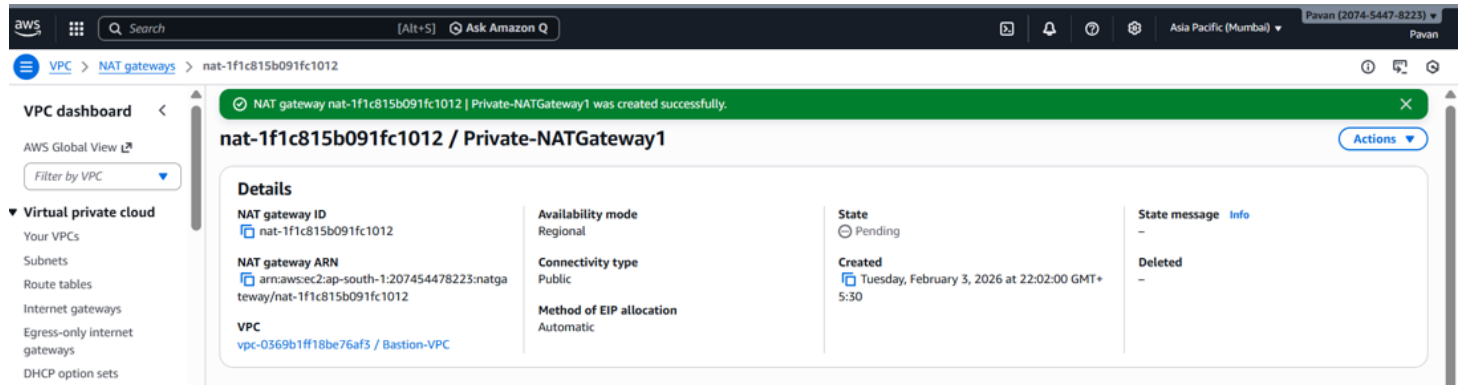
- Subnet Associations: Edit
- Select Bastion-Public-Subnet1
- Save

The screenshot shows the AWS Management Console interface for the 'Bastion-Public-RT1' route table (rtb-0ae4cc4cbfae21df9). The left sidebar shows the 'Virtual private cloud' section with 'Route tables' selected. The main content area displays the 'Subnet associations' tab, showing a table of explicit subnet associations. The table has columns for Name, Subnet ID, IPv4 CIDR, and IPv6 CIDR. One association is listed: 'Bastion-Public-Subnet1' with Subnet ID 'subnet-0e5b8f9ef22913a37' and IPv4 CIDR '10.0.1.0/24'. The 'IPv6 CIDR' column is empty.

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Bastion-Public-Subnet1	subnet-0e5b8f9ef22913a37	10.0.1.0/24	-

Phase 5: Create NAT Gateway

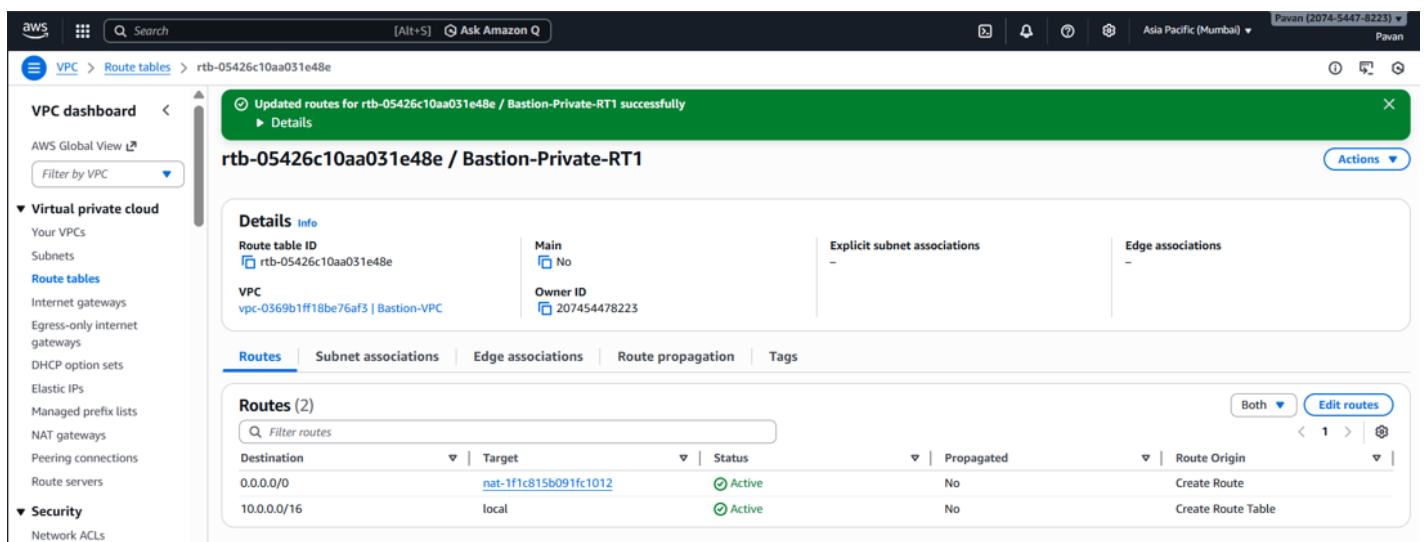
- Go to NAT Gateway → Create
- Name: Private-NATGateway1
- Subnet: Bastion-Public-Subnet1
- Connectivity type: Public
- Elastic IP: Allocate Elastic IP
- Click create NAT Gateway
- Wait until status becomes Available



Phase 6: Create Private Route Table:

Private Route Table:

- Go to Route Tables → Create
- Name: Bastion-Private-RT1
- VPC: Bastion-VPC
- Click create



Add NAT Route:

- Select Bastion-Private-RT1
- Routes: Edit routes
- Add:
 - Destination: 0.0.0.0/0
 - Target: NAT Gateway (Private NAT Gateway1)
- Save

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
0.0.0.0/0	igw-0609f1b7ed9cb7c86	-	-	-

Buttons: Add route, Cancel, Preview, Save changes

Associate Private Subnet:

- Subnet associations: Edit
- Select: Bastion-Private-Subnet1
- Save

rtb-05426c10aa031e48e / Bastion-Private-RT1

Details

- Route table ID: rtb-05426c10aa031e48e
- Main: No
- VPC: vpc-0369b1f18be76af3 | Bastion-VPC
- Owner ID: 207454478223

Subnet associations

Explicit subnet associations (0)

No subnet associations
You do not have any subnet associations.

Subnets without explicit associations (1)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Bastion-Private-Subnet1	subnet-0354da7d6a1109fc4	10.0.2.0/24	-

Buttons: Edit subnet associations

Phase 7: Create Security Group

Bastion Host Security Group:

- Go to Security Group → Create
- Name: Bastion-SG1
- VPC: Bastion-VPC
- Inbound Rule:
 - Type: SSH
 - Port Range: 22
 - Source: Anywhere IPv4 (0.0.0.0)

sg-06c11480a721efcb3 - Bastion-SG1

Details

Security group name Bastion-SG1	Security group ID sg-06c11480a721efcb3	Description SSH Access	VPC ID vpc-0369b1ff18be76af3
Owner 207454478223	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules (1/1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-009045280b21925d4	IPv4	SSH	TCP	22	122.168.70.147/32	-

- Outbound Rule:
 - Type: All traffic
 - Port Range: All
 - Destination: 0.0.0.0/0
- Click create

sg-06c11480a721efcb3 - Bastion-SG1

Details

Security group name Bastion-SG1	Security group ID sg-06c11480a721efcb3	Description SSH Access	VPC ID vpc-0369b1ff18be76af3
Owner 207454478223	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Outbound rules (1/1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Destination	Description
-	sgr-018a1713933fe3586	IPv4	All traffic	All	All	0.0.0.0/0	-

Phase 8: Create Private EC2 Security Group

Private EC2 Security Group:

- Create Security Group
- Name: EC2-Private-SG1
- VPC: Bastion-VPC
- Inbound Rule:
 - Type: SSH, All ICMP
 - Source: Bastion-SG1 (Bastion-Host)

sg-02eaf66c3fe56e717 - EC2-Private-SG1

Details

Security group name EC2-Private-SG1	Security group ID sg-02eaf66c3fe56e717	Description ec2 ssh access	VPC ID vpc-0369b1ff18be76af3
Owner 207454478223	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules (2/2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0740e41d1e74f9403	-	SSH	TCP	22	sg-06c11480a721efcb3...	-
-	sgr-0e21edae802273db6	-	All ICMP - IPv4	ICMP	All	sg-06c11480a721efcb3...	-

- Outbound Rule:
 - Type: All traffic
 - Destination: 0.0.0.0/0
- Click create

sg-02eaf66c3fe56e717 - EC2-Private-SG1

Details

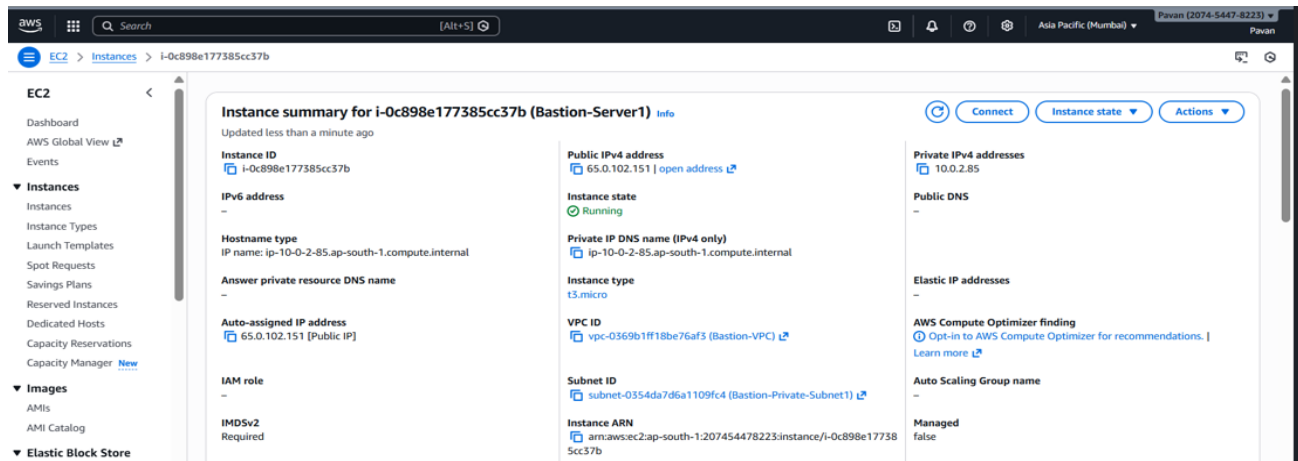
Security group name EC2-Private-SG1	Security group ID sg-02eaf66c3fe56e717	Description ec2 ssh access	VPC ID vpc-0369b1ff18be76af3
Owner 207454478223	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Outbound rules (1/1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Destination	Description
-	sgr-0bbdf85e24484d6f0	IPv4	All traffic	All	All	0.0.0.0/0	-

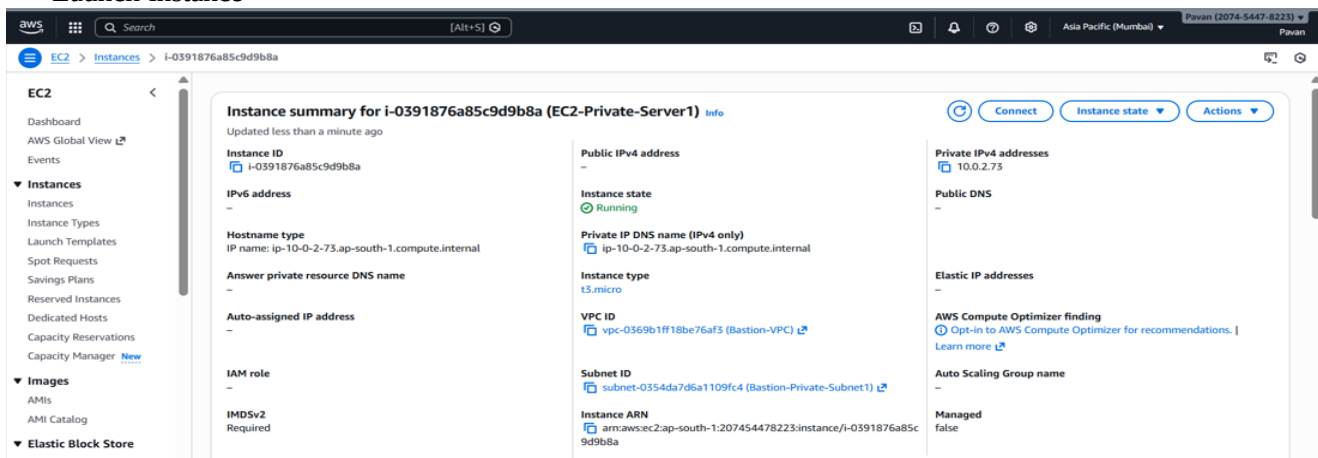
Phase 9: Launch Bastion Host (Public EC2 Instance)

- Go to EC2 → Launch instance
- Name: Bastion-Server1
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select/Create (Project-Key)
- Network Settings:
 - VPC: Bastion-VPC
 - Subnet: Bastion-Public-Subnet1-ap-south-1a
 - Auto-assign Public IP: Enable
 - Security Group: Create Bastion-SG1
- Launch instance



Phase 10: Launch Private EC2 Instance

- Click Launch instance
- Name: EC2-Private-Server1
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select Same Key (Project-Key)
- Network:
 - VPC: Bastion-VPC
 - Subnet: Bastion-Private-Subnet1
 - Auto-assign Public IP: Disable
 - Security Group: EC2-Private-SG1
- Launch instance



Phase 11: SSH Access (Bastion Host → Private Ec2 instance)

- **From Local Machine:** Used Command Prompt

[illegible]

- **From Bastion-Host:**

```

ec2-user@ip-10-0-2-73:~
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pavan>cd C:\Users\pavan\Downloads

C:\Users\pavan\Downloads>ssh -i project-Key.pem ec2-user@43.205.110.223

#_
_###_ Amazon Linux 2023
_#####
_###|
_#/--- https://aws.amazon.com/linux/amazon-linux-2023
_V~'
_/_/
_/_/

Last login: Tue Feb 3 16:56:55 2026 from 122.168.70.147
[ec2-user@ip-10-0-1-141 ~]$ ls
[ec2-user@ip-10-0-1-141 ~]$ ls
[ec2-user@ip-10-0-1-141 ~]$ exit
logout
Connection to 43.205.110.223 closed.

C:\Users\pavan\Downloads>scp -i project-Key.pem project-Key.pem ec2-user@43.205.110.223:/home/ec2-user/
project-Key.pem 100% 1678 20.2KB/s 00:00

C:\Users\pavan\Downloads>ssh -i project-Key.pem ec2-user@43.205.110.223

```

```
ec2-user@ip-10-0-2-73:~  
#_##### Amazon Linux 2023  
#\#####  
#\###|  
#\##/  
V~!!--> https://aws.amazon.com/linux/amazon-linux-2023  
  
./m/'!  
last login: Tue Feb 3 16:58:41 2026 from 122.168.70.147  
[ec2-user@ip-10-0-1-141 ~]$ ls  
project-Key.pem  
[ec2-user@ip-10-0-1-141 ~]$ chmod 400 project-Key.pem  
[ec2-user@ip-10-0-1-141 ~]$ ssh -i project-Key.pem ec2-user@10.0.2.73  
The authenticity of host '10.0.2.73 (10.0.2.73)' can't be established.  
ED25519 key fingerprint is SHA256:K2PWBqmUjJBbd960+Th7SHDA3PlQvzkUF0AlwBaAs/s.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.2.73' (ED25519) to the list of known hosts.  
  
#_##### Amazon Linux 2023  
#\#####  
#\###|  
#\##/  
V~!!--> https://aws.amazon.com/linux/amazon-linux-2023  
  
./m/'!  
[ec2-user@ip-10-0-2-73 ~]$ ping google.com  
PING google.com (142.250.206.110) 56(84) bytes of data
```

Troubleshooting:

Issue 1: SSH command failed due to missing or inaccessible “.pem” key file

Cause: Key file was not present in the Bastion Host.

- Incorrect file path used
- File permissions not set correctly

Solution: Copied the “.pem” file from the local machine to the Bastion Host securely using,

Cmd: “scp -I Project-Key.pem Project-Key.pem ec2-user@13.232.15.159: /home/ec2-user/”

- Verified the file location using “ls” command
- Set proper permissions using:

Cmd: “chmod 400 Project-Key.pem

Issue 2: Private EC2 responds to ping but SSh connection fails

Cause: ICMP (ping) allowed, but SSH (port 22) not allowed in security group

Solution: Added inbound SSH rule in private EC2 security group allowing traffic only from Bastion Host security group.

Conclusion

The custom VPC setup with Bastion Host and NAT Gateway successfully demonstrates the implementation of a secure and well-structured AWS network architecture. By segregating resources into public and private subnets, the project ensures strong network isolation and enhanced security. The Bastion Host acts as a controlled entry point for administrative access, while the NAT Gateway enables private EC2 instances to access the internet securely without being exposed to inbound traffic. Proper configuration of route tables and security groups ensures efficient traffic flow and adherence to AWS best practices. Overall, this project provides hands-on experience in designing, deploying, and managing a scalable and secure cloud infrastructure suitable for real-world production environments.

Issue 2: